

Sooyong Shin

📍 Daejeon, South Korea ✉ exceldra5@gmail.com in exceldra5 🌐 exceldra5

Research Interests

I am interested in **Computer Vision and Robot Perception**, particularly in building **Spatial Intelligence** that enables robots to understand and interact with complex real-world environments. My research interests span **SLAM**, geometric and semantic scene understanding, and temporally persistent perception. I am particularly interested in integrating **Multimodal and Vision-Language Models** with spatial representations and memory to connect visual perception with context-aware reasoning and reliable robotic action.

Keywords: Robot Perception, Spatial Intelligence, SLAM, Vision-Language Models

Education

Korea Advanced Institute of Science and Technology <i>B.S. Candidate in Electrical Engineering and Computer Science (Double Major)</i> <i>English: TOEFL iBT 5.5/6.0 (MyBest Scores)</i>	<i>Daejeon, South Korea</i> <i>Feb 2018 – Feb 2027</i>
---	---

Research Experience

NMAIL (Supervisor: Sungho Jo) <i>Research Intern</i> Conducted research on robot perception	<i>Daejeon, South Korea</i> <i>Mar 2026 – Jul 2026</i>
URL (Supervisor: Hyun Myung) <i>Research Intern</i> Conducted research on robot perception	<i>Daejeon, South Korea</i> <i>Jun 2025 – Feb 2026</i>
<i>Individual Study</i> Conducted an individual study on ROS and visual SLAM	<i>Mar 2025 – Jun 2025</i>
KC-ML2 <i>Research Intern</i> Co-first author of a paper currently under review at NeurIPS 2026 Evaluations & Datasets Track 🌐 - Conducted research on relocalization for robot mapping - Investigated text-based failure recovery in monocular visual SLAM 🌐 - Skills: C++, Python, ORB-SLAM3, YOLO, PaddleOCR	<i>Seoul, South Korea</i> <i>Aug 2024 – Feb 2025</i>
SIIT Lab (Supervisor: Junmo Kim) <i>Individual Study</i> Conducted an individual study on vision-language models	<i>Daejeon, South Korea</i> <i>Jul 2024 – Aug 2024</i>

Work Experience

Bookend <i>Software Engineer</i> - Built the frontend and backend of SENTENCIFY 🌐 (Chrome Web Store 🌐), a multilingual AI writing and paraphrasing application integrating multiple large language model APIs - Developed the service using React, FastAPI, and MongoDB	<i>Daejeon, South Korea</i> <i>May 2023 – Aug 2024</i>
<i>Software Engineering Intern</i> Developed TYPHERE, a React Native mobile platform connecting writers and publishers throughout the publishing workflow	<i>Dec 2022 – Feb 2023</i>

Projects

CareON — Safe Medical AI for Post-Discharge AML Care

Jul 2026

Excellence Award (2nd Place), Medical AI Competition 2026  

- Served as **AI & Systems**, designing a fail-closed medical AI system for post-discharge AML care.
- Integrated deterministic safety rules, grounded RAG, abstention, and audit logging to support reliable and traceable patient guidance.
- Built the **FastAPI** backend and **Flutter** client, and evaluated the system using synthetic clinical scenarios.

KAIST Swimming Team KAORI Web Platform www.kaist-kaori.org

Since 2024

- Developed and maintained a full-stack web platform for managing member profiles and swimming records
- Enabled individual and team performance analysis through structured historical record data
- Skills: React, FastAPI, MongoDB

Skills

Programming: Python, C++, JavaScript, TypeScript

AI & Robotics: PyTorch, ROS, ORB-SLAM3, YOLO, PaddleOCR

Development: React, React Native, FastAPI, MongoDB, Git

Leadership Experience

KAIST MR (Microbot Research)

Daejeon, South Korea

Member

Since Mar 2025

AI Robotics KR

South Korea

Community Manager

Oct 2024 – Sep 2025

Community Lead

Sep 2025 – Present

Republic of Korea Army

Paju, South Korea

Sergeant; honorably discharged

Mar 2019 – Nov 2020

Korean Red Cross Lifeguard

Daejeon, South Korea

Since Aug 2018

KAIST Swimming Team KAORI

Daejeon, South Korea

Senior Member

Since Mar 2018

President of KAORI

2022

Chair, KAIST Swimming Competition

2023